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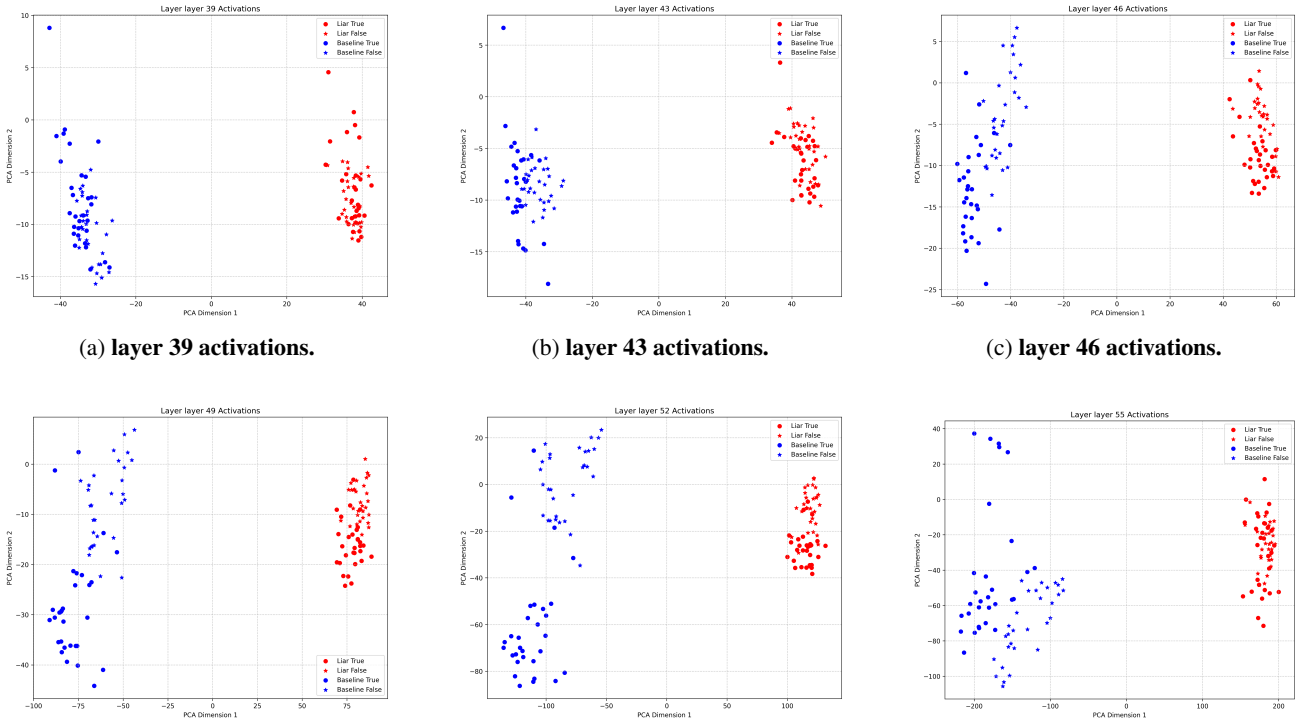
A. Appendix / Supplemental Material

A.1. PCA Visualization of Residual Activations



(a) **layer 39 activations.** (b) **layer 43 activations.** (c) **layer 46 activations.**

(d) **layer 49 activations.** (e) **layer 52 activations.** (f) **layer 55 activations.**

Figure 4: **The visualization of residual stream activations following Principal Component Analysis (PCA).** Red markers denote the residual stream activations for template $\mathcal{T}_a$ (designated as liar condition), while blue markers represent vectors for template $\mathcal{T}_b$ (baseline condition). Circular markers indicate stimuli s where the presented statement was true, and star-shaped markers correspond to false statements.

Our experimental framework for experiment 1 can categorize data points into four distinct classes based on template type and statement veracity for each layer:

- Template A: True statements $a_{T,i}$ and false statements $a_{F,i}$
- Template B: True statements $b_{T,i}$ and false statements $b_{F,i}$

After performing PCA dimensionality reduction to 2D space, we observe intriguing dynamics in the model's representation space. Previous studies (Yang & Buzsaki, 2025; Campbell et al., 2023) examining instructed deception (different with our Template A) identified three characteristic phases: (1) Early layers exhibit separation between $\{a_{T,i}, a_{F,i}\}$ and $\{b_{T,i}, b_{F,i}\}$ clusters; (2) Middle layers show differentiation between true and false statements within each template (a_T vs. a_F and b_T vs. b_F); (3) Late layers demonstrate an initial alignment followed by gradual reversal of the difference vectors $\bar{a}_T - \bar{a}_F$ and $\bar{b}_T - \bar{b}_F$, ultimately approaching antiparallel orientation.

However, our results (Figure 4) reveal a more complex trajectory. The first phase persists until layers 43-44, followed by a second phase extending to layers 51-52. Contrary to previous findings, we observe no subsequent reversal in later layers. Notably, while b_T and b_F representations remain separable, the a_T and a_F clusters exhibit gradual reconvergence in the final layers. This suggests that our threat lying template, Template $\mathcal{T}_a$, indeed differs in semantic content from the instructed lying templates examined in prior studies. This distinction manifests in the representational differences observed within the residual stream space. The gradual reconvergence of clusters in the final layers may reflect the model's internal conflict or moral dilemma during deceptive behavior generation.